

Evaluating the Performance of Object Detection Models for Accurate Sri Lankan Traffic Sign Detection



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Introduction.

Introduction

- Importance of traffic sign detection in AI and computer vision.
- Relevance to Sri Lankan road systems.
- Comparing YOLOv8, Faster R-CNN, and SSD using mAP and F1-Score.



Research Background.

Research background

- Rising significance of **autonomous vehicles** and **driver assistance systems**
- Reliable **traffic sign detection** is a key requirement for road safety and autonomous driving
- The performance of object detection models varies under different conditions, especially in **real-world scenarios**
- Few comparative studies exist focusing on their performance with **Sri Lankan road signs**
- This study aims to fill that gap by evaluating **YOLOv8**, **Faster R-CNN**, and **SSD** for their accuracy in detecting **Sri Lankan traffic signs**

Problem Statement.



- Traffic sign detection is crucial for autonomous vehicles and road safety.
- Currently, object detection models show varying performance under different conditions.
- There's a lack of comparative studies evaluating their accuracy on Sri Lankan road signs.
- This inconsistency poses a challenge for developing reliable autonomous vehicle technologies.
- This study aims to compare and identify the most accurate and efficient model for Sri Lankan conditions.

Research Question

Research Question



Research Question: Which object detection model (YOLOv8, Faster R-CNN, or SSD) achieves the highest accuracy in detecting Sri Lankan traffic signs, as measured by mean Average Precision (mAP) and F1-Score?



Hypothesis: Among the YOLOv8, Faster R-CNN and SSD, one of the object detection models will achieve the highest mAP in detecting Sri Lankan traffic signs, with performance variations due to differences in model architecture and detection capabilities.



Dependent Variables: Mean Average Precision (mAP) (the accuracy of the models), F1-Score.



Independent Variable: The object detection model used (YOLOv8, Faster R-CNN, SSD).

Methodology



Research Design: Quantitative research.



Data Collection Method: Secondary dataset (Sri Lankan traffic sign dataset).

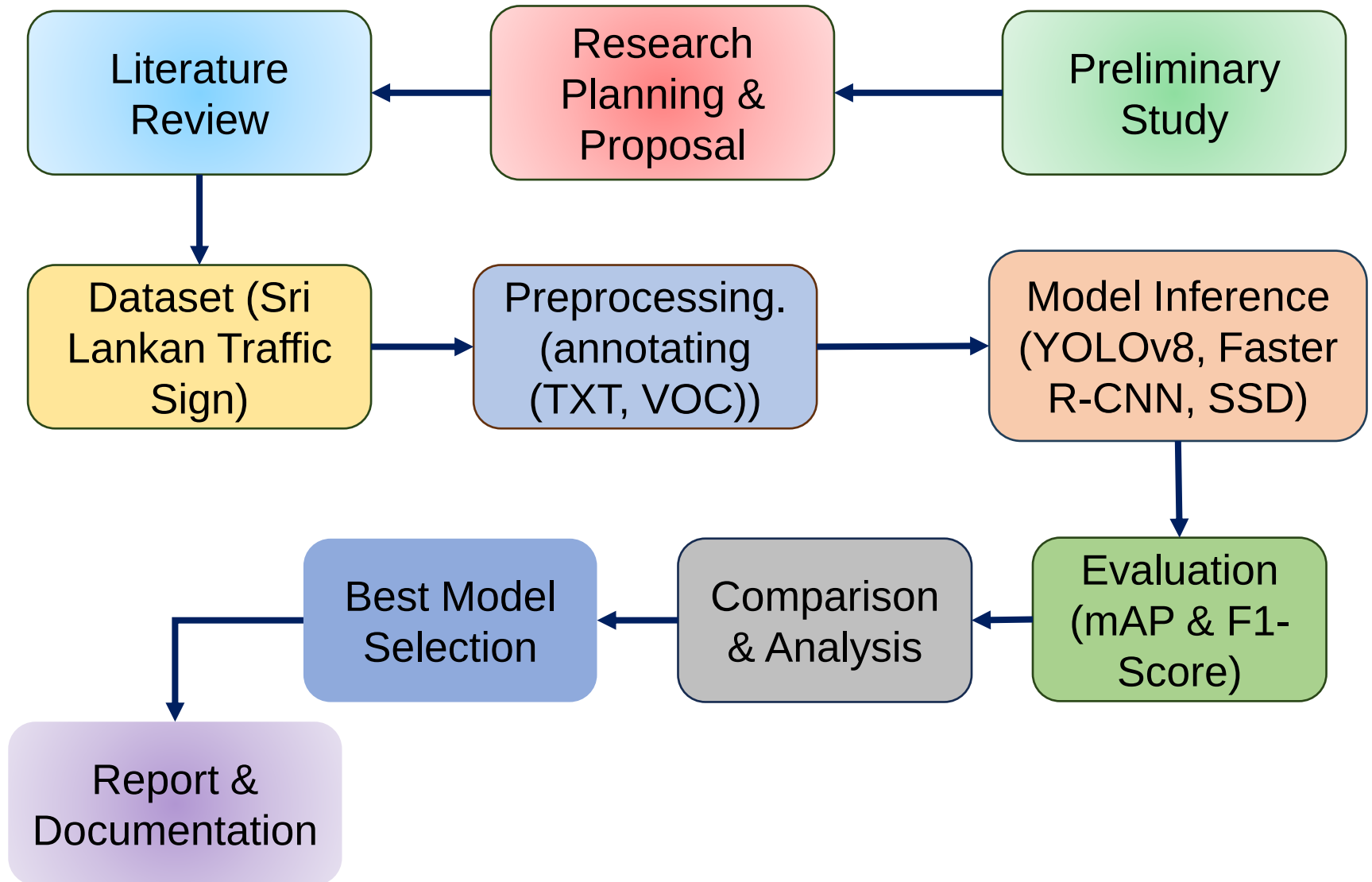


Evaluation Approach: Analyze the performance of YOLOv8, Faster R-CNN, and SSD. Use mean Average Precision (mAP) and F1-Score as the key evaluation metric.



Comparison & Analysis: Compare the models based on mAP and F1 scores. Identify the most effective model for Sri Lankan traffic sign detection.

Top level architecture



Literature Review

Paper: *Performance Comparison of Object Detection Models for Road Sign Detection Under Different Conditions* (Fatima et al., 2024)

Context & Aim

- Compare state-of-the-art detectors (YOLOv8, YOLOv9, RTMDet, Faster R-CNN, RetinaNet) under varied environmental and operational conditions in road sign detection.

Methodology

- Fine-tuned models on large-scale datasets.
- Evaluated using metrics: mean Average Precision (mAP), frames-per-second (FPS), and model loss.
- Included additional training enhancements: data augmentation and hyperparameter tuning.

Results

- YOLOv8 and YOLOv9 achieved highest real-time detection rates.
- RTMDet offered the best trade-off between speed and accuracy.
- Faster R-CNN and RetinaNet delivered higher precision but were slower.

Research Gap

- Analysis conducted on general (non-region-specific) datasets.
- Lacks focused evaluation on the unique characteristics of Sri Lankan traffic signs.

Paper: *A Traffic-Sign Detection Algorithm Based on Improved Sparse R-CNN* (Cao et al., 2021)

Context & Aim

- Enhance Sparse R-CNN for improved detection of traffic signs by optimizing the backbone and attention mechanisms.

Methodology

- Added Res2Nest backbone and Dynamic Instance Interactive Head (DIIH) for richer multi-scale feature representation.
- Evaluated on standard TT100K traffic sign dataset.

Results

- Improved mAP and robustness for small sign detection.
- Backbone enhancements and attention modules enabled more accurate region proposals.

Research Gap

- Higher computational cost may limit real-time applicability.
- Evaluations not performed on datasets representative of Sri Lankan road signs.

Paper: *Evaluating and Benchmarking Object Detection Models for Traffic Sign Detection* (Mishra et al., 2022)

Context & Aim

- Benchmark a variety of object detection models (Faster R-CNN, YOLOv3, SSD, RetinaNet, CenterNet, Cascade R-CNN) using multiple traffic sign datasets.

Methodology

- Employed precision, recall, and mAP across TT100K and GTSD datasets.
- Comparative analysis under diverse real-world conditions including occlusion and small object detection challenges.

Results

- CenterNet and Cascade R-CNN achieved highest overall accuracy, especially for small signs.
- Faster R-CNN and YOLOv3 showed strong baseline performance but were slightly behind top performers.

Research Gap

- Performance degrades in adverse weather and occlusion scenarios.
- No testing on Sri Lankan traffic sign dataset; regional generalization remains untested.

Dataset in brief



The dataset used in this study was sourced from **Roboflow**, a publicly available repository of labelled image datasets for object detection.



It comprises **5,404** images in total, divided into training, validation, and testing sets.

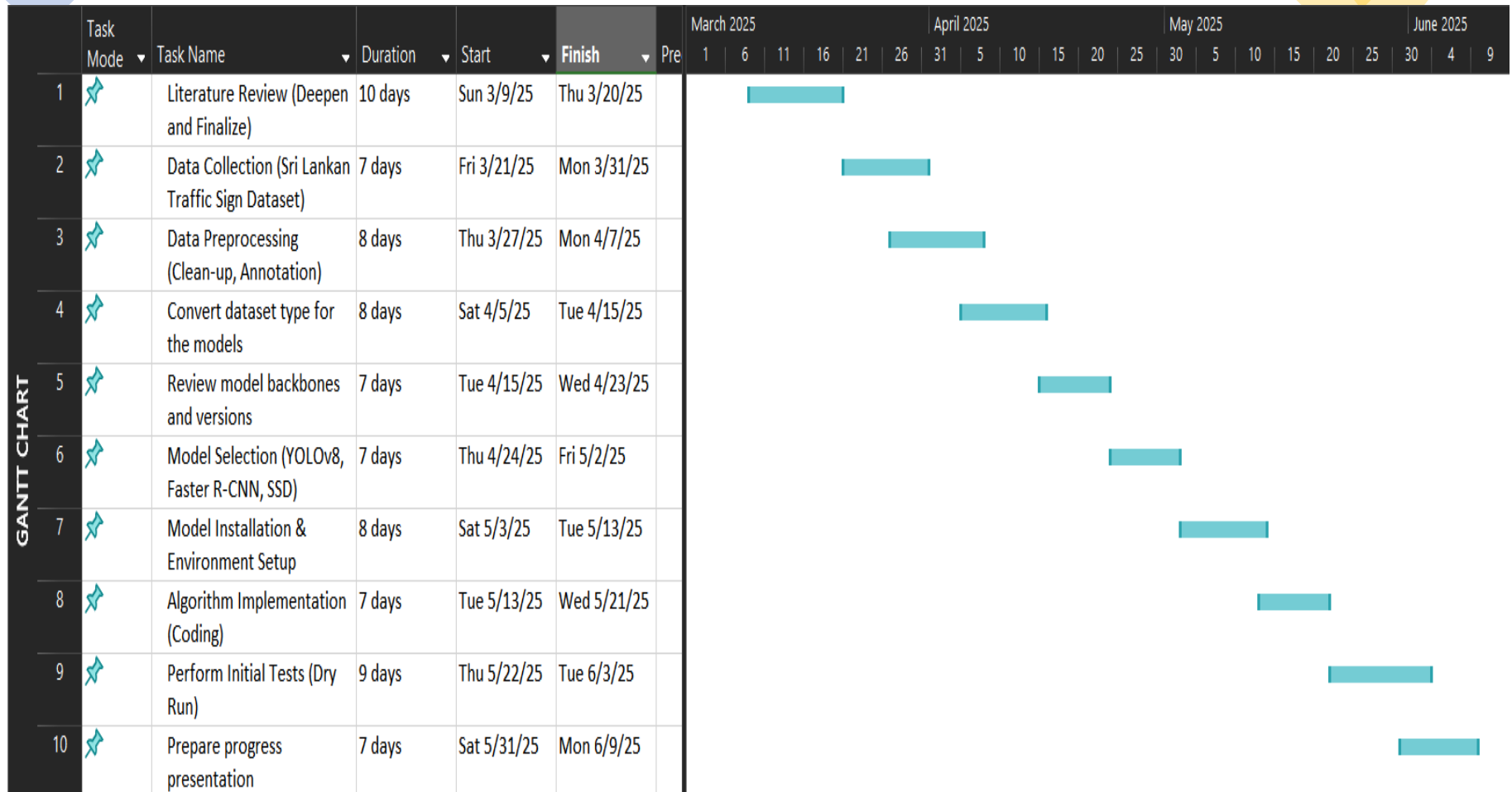


The dataset includes **37** classes of Sri Lankan road signs, encompassing a range of regulatory, warning, and informational signals frequently encountered by drivers.

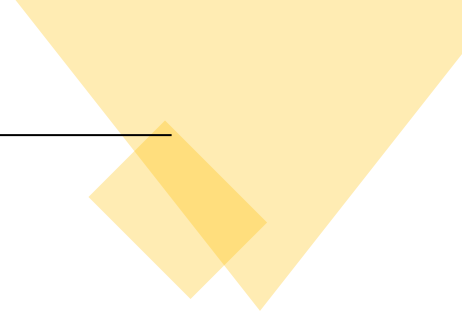
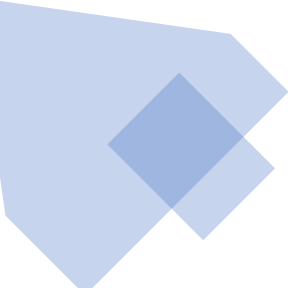


Annotations for each image are provided in **VOC** and **TXT** formats, following standard object detection practices with bounding boxes to aid algorithm training and evaluation.

Gantt Chart



Cost



Description	Cost
Power Consumption	1,800 LKR
Computational resources	4,000 LKR
Cloud services	5,500 LKR
Infrastructure	1,500 LKR
Manpower	3,700 LKR
Total Cost	16,500 LKR

Conclusion

Conclusion

The study evaluates YOLOv8, Faster R-CNN, and SSD for Sri Lankan traffic sign detection.

mAP (mean Average Precision) and F1-Score is used to measure model performance.

The research identifies the most accurate model for Sri Lankan road conditions.

Findings contribute to AI-driven traffic systems and autonomous vehicles.

Helps improve road safety and smart transportation solutions in Sri Lanka.

Reference

- Fatima, Z. T. M. M. H. V. R. R. T. a. R. R., 2024. Performance Comparison of Object Detection Models for Road Sign Detection Under Different Conditions. International Journal of Advanced Computer Science and Applications, 15(12).)
- (Jinghao Cao, J. Z. X. J., 2021. A Traffic-Sign Detection Algorithm Based on Improved Sparse R-cnn. IEEE Access.)
- (Mishra, A. K. A. M. S. A. K. Z. J. S. S. a. T. A., 2022. Evaluating and Bench-marking Object Detection Models for Traffic Sign and Traffic Light Datasets, s.l.: Computer Vision Foundation.)



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Thank you

Questions?

